

Neutrino Isocurvature and the S8 Power Damping

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Abstract. We solve the cosmic matter clustering tension (S8) by activating mixed neutrino-isocurvature initial conditions. We show that freezing the neutrino isocurvature fraction at the topological value $\beta_{iso} = 1/36$ *dampsthematterpowerspectrumonsmallscales*

1 Introduction

The S8 tension measures the discrepancy in matter clustering between CMB and weak lensing surveys. We resolve this using neutrino isocurvature.

2 Theoretical Formulation

Here we formulate the core mathematical relations governing the system:

$$\beta_{iso} = \frac{1}{36} \approx 0.027778 \quad (\text{isocurvature power fraction}) \quad (1)$$

$$S_8 = \sigma_8 \sqrt{\frac{\Omega_m}{0.3}} \approx 0.791 \quad (\text{matter clustering amplitude}) \quad (2)$$

3 Isocurvature Power Damping

The isocurvature fraction suppresses the growth of structure on small scales during the radiation-to-matter transition, lowering S8 without changing Planck's H_0 .

4 Conclusion

The isocurvature fraction $\beta_{iso} = 1/36$ *resolves the S8 tension, providing a unified explanation for late-time structure growth.*

References

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